Machine Learning for Mix Design of Concrete

B-Tech Project – 1

Submitted by

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CERTIFICATE

This is to certify that the project report entitled "**Machine Learning for Mix Design of Concrete**" submitted by **Rudra Sai Teja, Roll No. 21CE31010** to Indian Institute of Technology Kharagpur is a record of bona fide project work carried out under my supervision and guidance for the course BTech Project - 1 (CE47005) in the Department of Civil Engineering at the Indian Institute of Technology Kharagpur.

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Date:
Place: IIT Kharagpur

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I hereby declare that

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1. INTRODUCTION

1.1 Background

Concrete mix design is one of the critical aspects in civil engineering and construction engineering. The main motive of this process is to ensure that the concrete meets the required strength, workability and durability criteria for specific applications. Traditionally, mix design involves empirical methods guided by standard codes such as IS 10262:2019, ACI 211.1, and BS EN 206. These approaches, while reliable, can be time-consuming and may not account for the complexities introduced by modern construction demands and the growing variety of available materials.

The process involves determining the optimal proportions of ingredients—cement, fine aggregate, coarse aggregate, water, and admixtures—based on criteria mentioned above. The codes, above mentioned, offer step-by-step procedures derived from decades of experimental research and engineering expertise. While these approaches are reliable and widely adopted, they often depend on trial-and-error adjustments and heuristic judgment to fine-tune mix proportions.

Due to this iterative process, it consumes a lot of resources, manpower and time. Also due to the evolution of specifications of the construction industry and their demands, this process has added many layers of complexity such as sustainability, etc.

Due to these alternative procedures have come into the picture to overcome the limitations of the traditional procedure.

1.2 Applications of Machine Learning in Concrete Mix

The availability of data of the mix proportions and corresponding compressive strength the application of machine learning in this area to predict the mix proportion for the specified grade of concrete could overcome the limitations such as wastage of the resources, manpower, etc.

However, this application can only replace the prediction of the mix proportion of the concrete. Before applying it in the field the results predicted by machine learning need to be evaluated in the lab by performing in the lab to ensure the variability of the compressive strength of the concrete is within the range of acceptable limits. The tests such as batching and casting of concrete, compressive strength of concrete by destructive techniques, and workability all such tests need to be performed before adopting the results of the machine learning model completely.

1.3 Objectives and Scope

The project's objective is to develop and analyse a machine learning model to predict the concrete composition for different grades of concrete and optimise it to an optimal level and further create a User Interface to make it accessible for everyone.

2. LITERATURE REVIEW

Concrete mix design is a systematic process that involves the selection and proportioning of materials to achieve the desired strength, durability, and workability for specific applications. This process is critical for ensuring that concrete structures meet performance requirements while maintaining cost-efficiency and sustainability.

The important components of concrete are:

- **Cement**

It acts as a primary binder in concrete. Its role is to hold the aggregates together. Ordinary Portland Cement (OPC) is the most commonly used type, though blended cements such as Portland Pozzolana Cement (PPC) and Portland Slag Cement (PSC) are also widely employed for their enhanced durability and environmental benefits.

- **Fine Aggregates**

Fine aggregate, typically sand, fills voids between coarse aggregates and contributes to the workability of the mix. River sand or manufactured sand (M-sand) is commonly used, with the choice depending on availability and the requirements of the mix.

- **Coarse Aggregates**

Coarse aggregates provide the bulk of the concrete mass and contribute to its strength and dimensional stability. Crushed stone or gravel is used as coarse aggregate, and its size, shape, and gradation significantly influence the concrete's properties.

- **Water**

Water is a vital component in concrete as it hydrates the cement to form the binding gel, giving concrete its strength. The water-cement ratio (w/c) is a critical factor influencing both strength and durability; excessive water reduces strength, while insufficient water hampers workability.

- **Additional Ingredients or Admixtures**

These are used in concrete preparation to modify the properties of the concrete mix, making it more suitable for specific applications. Their applications primarily lie in the improvement in the workability, durability, strength, and setting time of the concrete, as well as control factors like shrinkage and temperature changes. These admixtures are Supplementary Cementitious Materials such as fly ash, blast furnace slag and chemical admixtures like superplasticizers which improve workability of a concrete for a particular water-cement(w/c) ratio and water reducing admixtures which reduce the water-cement(w/c) ratio for a particular workability.

Group	Grade designation	Characteristics compressive strength of 150 mm cube at 28 days, N/mm2
Ordinary Concrete	M10 M15 M20	10 15 20
Standard Concrete	M25 M30 M35 M40 M45 M50 M55	25 30 35 40 45 50 55
High Strength Concrete	M60 M65 M70 M75 M80	60 65 70 75 80

Table 2.1 Compressive Strength of different grades of concrete

In this representation, "M" stands for Mix, and the number following "M" indicates the compressive strength of concrete in megapascals (MPa) according to IS 10262.

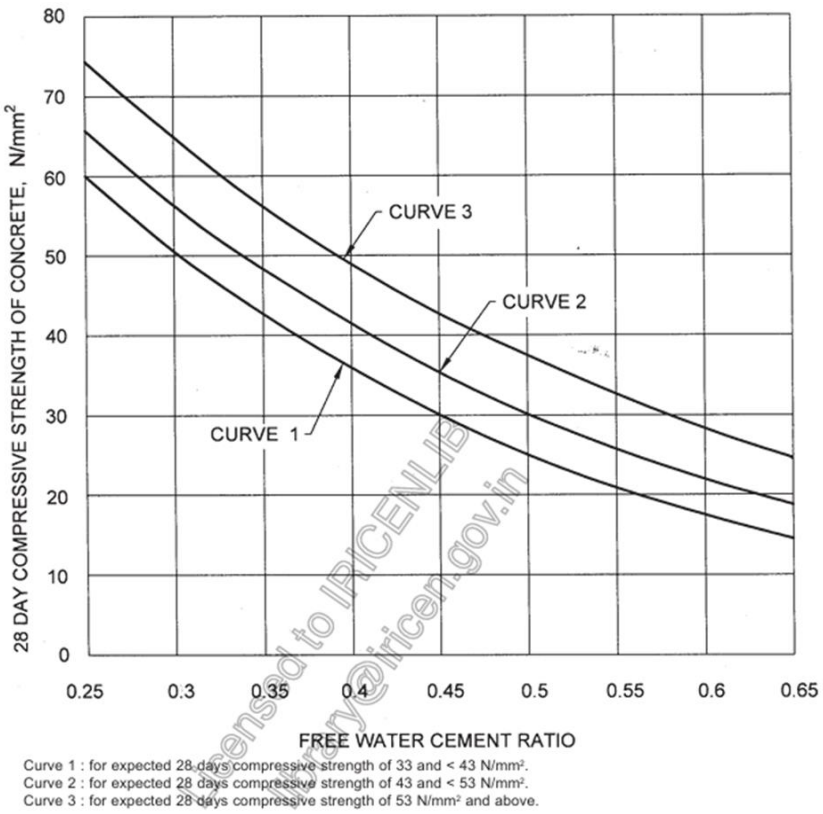

Figure 2.1 W/C ratio vs 28-day Compressive Strength of concrete for different grades of cement

In the traditional method, the first step is to estimate the water-cement(w/c) ratio. The graph above shows the relation between w/c ratio and the compressive strength. Based on the type of cement employed in the construction we can get the w/c ratio.

Sl No.	Nominal Max size of the aggregates (mm)	Water Content (kg)
1	10	208
2	20	186
3	40	165

Table 2.2 Water required for different nominal max size of the aggregates

Then the water content is estimated using the nominal maximum size of the aggregates. The table above provides the details of it.
Followed by this we estimate the cement content using water content and w/c ratio

Sl No.	Nominal Max size of the aggregates (mm)	Entrapped Air, as Percentage of Volume of Concrete
1	10	1.5
2	20	1.0
3	40	0.8

Table 2.3 Entrapped air for different nominal max size of the aggregates

The table above is used in estimating the air content in the concrete. The air content of the concrete is important because it affects the workability, durability and strength of the concrete. They further help during the shrinkage of concrete.

Table 5 Volume of Coarse Aggregate per Unit Volume of Total Aggregate for Different Zones of Fine Aggregate for Water-Cement/Water-Cementitious Materials Ratio of 0.50

(*Clause* 5.5)

Sl No.	Nominal Maximum Size of Aggregate mm	Volume of Coarse Aggregate per Unit Volume of Total Aggregate for Different Zones of Fine Aggregate			
		Zone IV	Zone III	Zone II	Zone I
(1)	(2)	(3)	(4)	(5)	(6)
i)	10	0.54	0.52	0.50	0.48
ii)	20	0.66	0.64	0.62	0.60
iii)	40	0.73	0.72	0.71	0.69

NOTES

1 Volumes are based on aggregates in saturated surface dry condition.

2 These volumes are for crushed (angular) aggregate and suitable adjustments may be made for other shape of aggregate.

3 Suitable adjustments may also be made for fine aggregate from other than natural sources, normally, crushed sand or mixed sand may need lesser fine aggregate content. In that case, the coarse aggregate volume shall be suitably increased.

4 It is recommended that fine aggregate conforming to Grading Zone IV, as per IS 383 shall not be used in reinforced concrete unless tests have been made to ascertain the suitability of proposed mix proportions.

Table 2.4 Ratio of coarse aggregates to the total aggregates

The table above gives the ratio of coarse aggregates required to the total aggregates in the preparation of concrete. Here the zones are classified on the basis of fine aggregates that pass through the 600µ sieve. The criteria for the different zones of fine aggregates are mentioned in the table in the following page.

Zone	Percentage
Zone I	15% to 34%
Zone II	34% to 59%
Zone III	60% to 79%
Zone IV	80% to 100%

Table 2.5 Criteria for different zones of fine aggregates

This is done on the basis percentage passing through 600µ sieve.
The aggregates required are calculated using the formula given below

$V = [W + C/Sc + 1/p \times fa/S_{fa}] \times 1/1000$ for fine aggregate 1

$V = [W + C/Sc + 1/(1-p) \times Ca/S_{ca}] \times 1/1000$ for coarse aggregate 2

Here W represents the water content, C represents the cement content, Sc represents the specific gravity of the cement, p represents the value derived from table 5 of IS 10262:2019 mentioned above, fa and ca represents fine and coarse aggregates S_{fa} and S_{ca} represents their specific gravities

After estimation of all the ingredients trial mixes and batching is done to check for the water, w/c ratios. Usually, 4 trial mixes are measured where the 1st trial mix gives the information on the water requirements and the following 3 mixes gives the information on the compressive strength and its relation with w/c ratio. Since all this process requires huge resources, manpower and time, we can use the machine learning techniques to avoid this wastage to some extent.

Patryk Ziolkowski and Maciej Niedostatkiewicz's 2019 paper "Machine Learning Techniques in Concrete Mix Design", introduces the use of Artificial Neural Networks to develop a practical tool for concrete mix design in engineering practice. The key goal was to predict the compressive strength of concrete based on the mix composition and provide optimal ingredient ratios to achieve desired strength levels. Machine learning (ML) is increasingly employed to address challenges in predicting and optimizing concrete properties. ML leverages advanced computational techniques to identify patterns and relationships within data that traditional methods may overlook. It also demonstrates that conventional mix design methods, though reliable, are labour-intensive and limited in handling modern complexities. Machine learning, particularly ANN and Deep Neural Networks (DNN), offers innovative solutions for predicting and optimizing concrete properties, especially compressive strength. However, practical implementation requires addressing challenges like data quality, computational efficiency, and validation of results through real-world testing.

ANNs are a popular ML technique for modelling complex relationships. Modelled after human brain neurons, they consist of input, hidden, and output layers. Input data passes through hidden layers, where patterns are identified, and predictions are generated. Features such as weight adjustments and multi-layered architecture enhance ANN's predictive capabilities.

ML, particularly ANN, has been widely used to predict the compressive strength of concrete. Early studies, such as Yeh et al. (1998), utilized ANN and linear regression to predict strength but faced challenges due to immature data inputs. Subsequent researchers improved methodologies, introducing innovations like modular networks, neural-expert systems, and feature scaling. DNNs offer higher generalization and precision, particularly for complex datasets, though they may suffer from computational expenses and potential underfitting in smaller datasets. We can also use SoftMax regression and feature scaling to enhance prediction accuracy by focusing on input ratios instead of raw quantities.

3. METHODOLOGY

The machine learning models employed in this project to predict the results are Lasso Regression, Ridge Regression, Linear Regression, Decision Tree Regressor, Random Forest Regressor

In machine learning, regression refers to a type of supervised learning where the goal is to predict a continuous output variable(dependent) based on one or more input variables (independent variables). The key characteristics of regression in a machine learning model are continuous output variable, input features, objective of the model and the evaluation metrics.

The objective is to learn a mapping function from inputs X to the target y:

$$y=f(X)+\epsilon$$

where ϵ represents the error term.

The evaluation metrics are

Mean Absolute Error (MAE): Average of absolute differences between predicted and actual values.

Mean Squared Error (MSE): Average of squared differences.

Root Mean Squared Error (RMSE): Square root of MSE for interpretability in the same unit as the target.

R-squared (R^2): Indicates how well the model explains the variability of the target.

- **Linear Regression**

Linear regression is a statistical method that models the relationship between a dependent variable (target) and one or more independent variables (predictors) by fitting a linear equation to observed data. The relationship is typically represented as

$$y=\beta_0 + \beta_1x_1 + \beta_2x_2 + \cdots + \beta_nx_n +\epsilon$$

where y is the target variable, x_1, x_2, …, x_n are predictor variables, β_0 is the intercept, and β_1, β_2,…,β_n are the coefficients.

It assumes a linear relationship between the predictors and the target, and that the residuals (errors) are normally distributed with constant variance.

- **Lasso Regression**

Lasso regression is a variation of linear regression that includes an L1 regularization term. This penalization helps in feature selection by shrinking less important feature coefficients to zero. The loss function is modified by adding a penalty term

$$\text{Loss function} = \text{Sum of squared residuals} + \lambda*\sum*|\beta|$$

where λ is the regularization parameter that controls the degree of shrinkage.

- **Ridge Regression**

Ridge regression is another regularization technique similar to Lasso, but it uses L2 regularization instead of L1. This means it penalizes the square of the coefficients rather than their absolute values. The loss function is modified as

$$\text{Loss function} = \text{Sum of squared residuals} + \lambda * \sum \beta^2$$

where λ is the regularization parameter.

- **Decision Tree Regressor**

A decision tree regressor is a non-linear regression model that splits the data into subsets based on feature values, using rules that partition the data in a hierarchical manner. The model makes predictions by averaging the target values in the leaves. The algorithm recursively splits the data into smaller groups, minimizing the variance in each group. The process stops when further splits do not reduce variance or when a stopping criterion is reached (e.g., a maximum depth). It can model non-linear relationships and handle complex interactions between features. It is interpretable, and you can visualize the decision-making process.

- **Random Forest Regressor**

A random forest regressor is an ensemble method that combines multiple decision trees to make predictions. It constructs many trees and aggregates their outputs (typically by averaging) to improve prediction accuracy and robustness. Each tree in the forest is trained on a random subset of the data and uses a random subset of features for splitting at each node. This randomness helps reduce overfitting compared to a single decision tree. It improves accuracy by reducing the variance of individual trees and is more robust than a single decision tree. It can handle high-dimensional data, missing values, and complex interactions.

4. ANALYSIS AND RESULTS

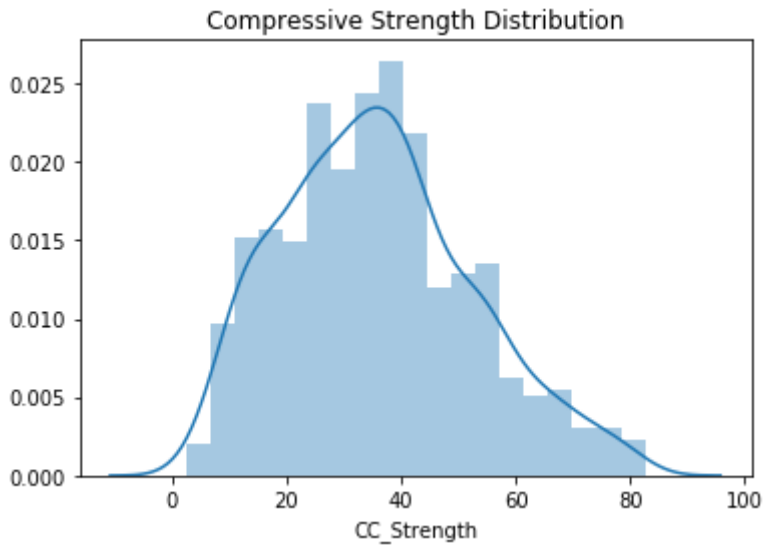

Figure 4.0.1 Distribution of the Compressive Strength of Concrete dataset

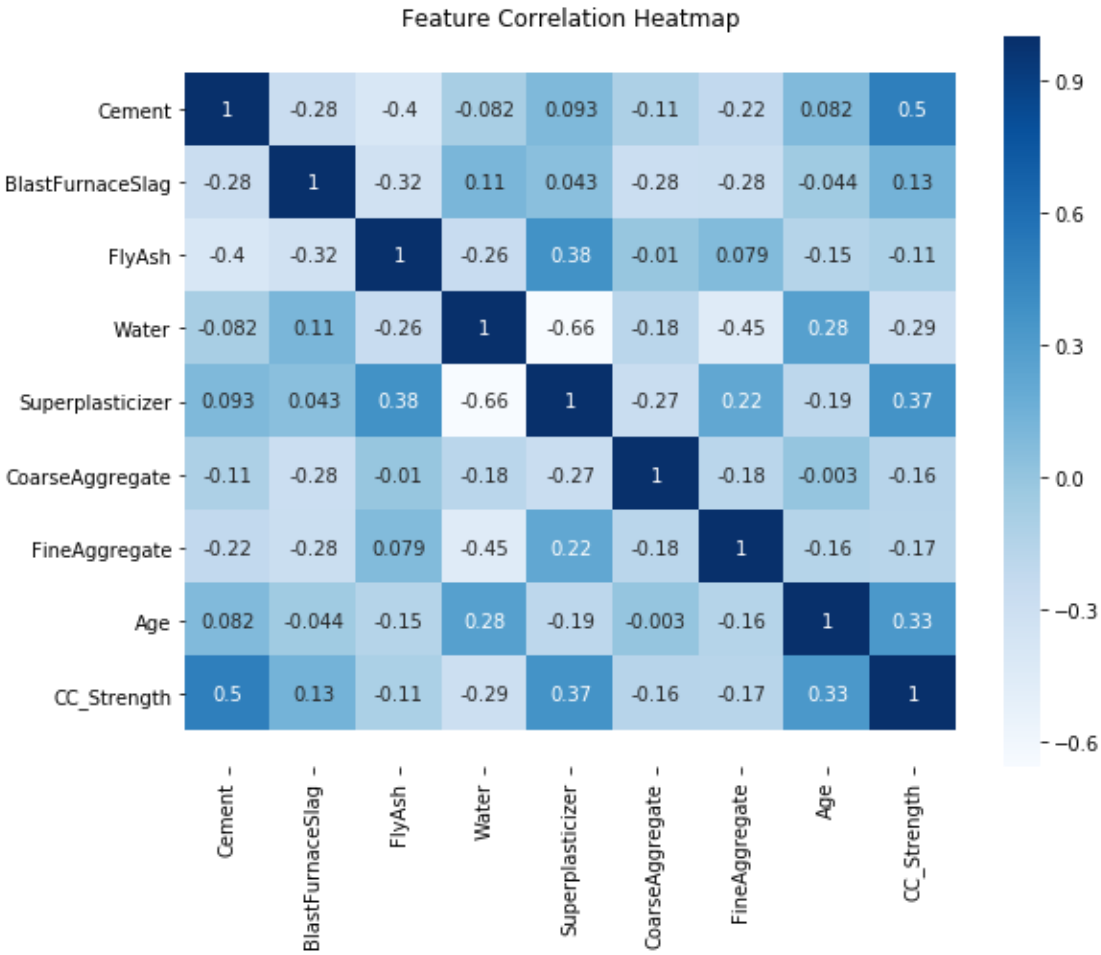

Figure 4.0.2 Correlation Heatmap of all the features

This chart represents the correlation between different parameters used for evaluation. Correlation matrix contains the correlations between all pairs of variables in a dataset. Correlation is a normalized measure of covariance, giving a standardized value that ranges from -1 to 1.

$$\mathrm{Corr}(X,Y) = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \cdot \sigma_Y}$$

where σ_X and σ_Y are the standard deviations of X and Y

This matrix helps in identifying the linear relationship between the variables. Also helps in further exploring the hidden patterns, multicollinear relationship, feature selection and improving the model performance.

4.1 Linear Regression

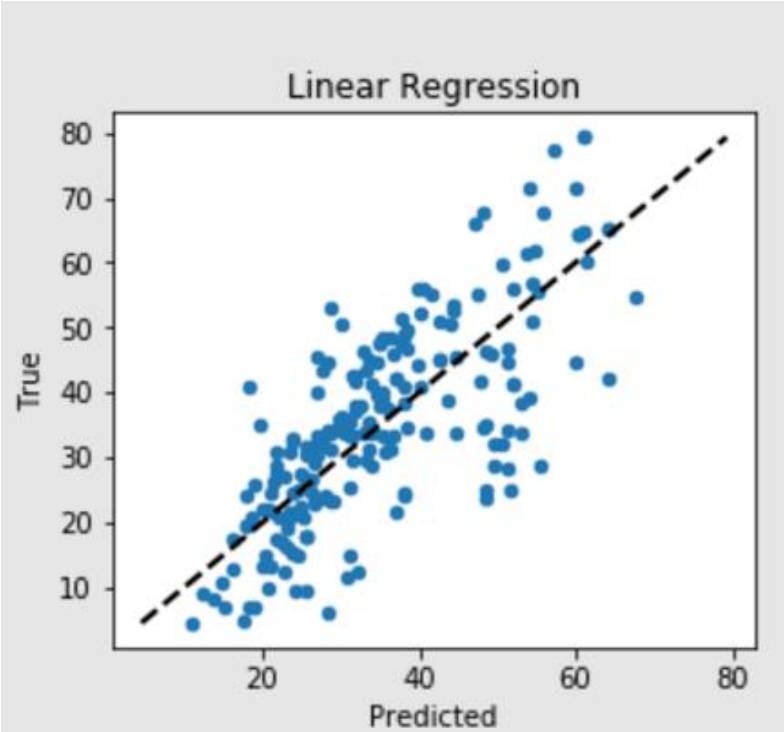

Figure 4.1.1 True vs Predicted values of Linear Regression

Model	RMSE	MSE	MAE	R^2
Linear Regression	10.29	105.78	8.23	0.57

4.2 Lasso Regression

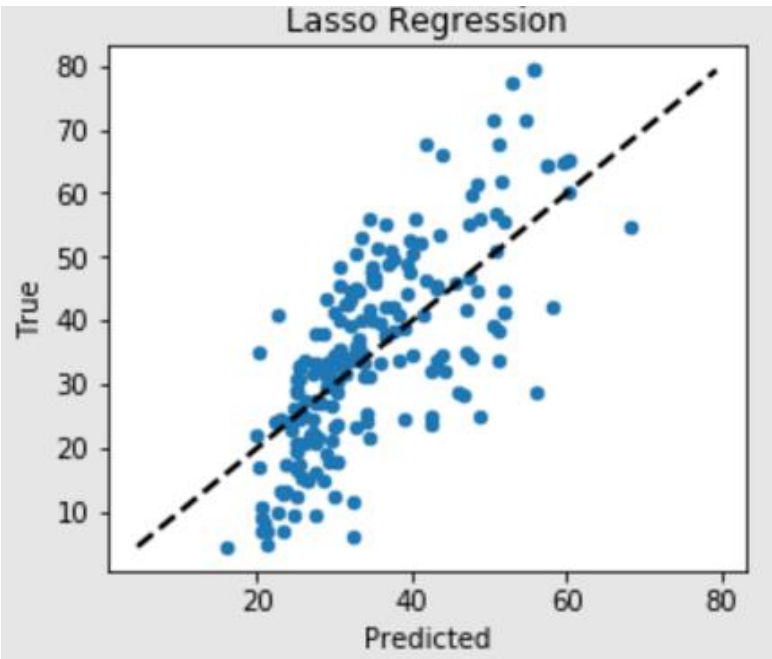

Figure 4.2.1 True vs Predicted values of Lasso Regression

Model	RMSE	MSE	MAE	R^2
Lasso Regression	10.68	114.13	8.66	0.54

4.3 Ridge Regression

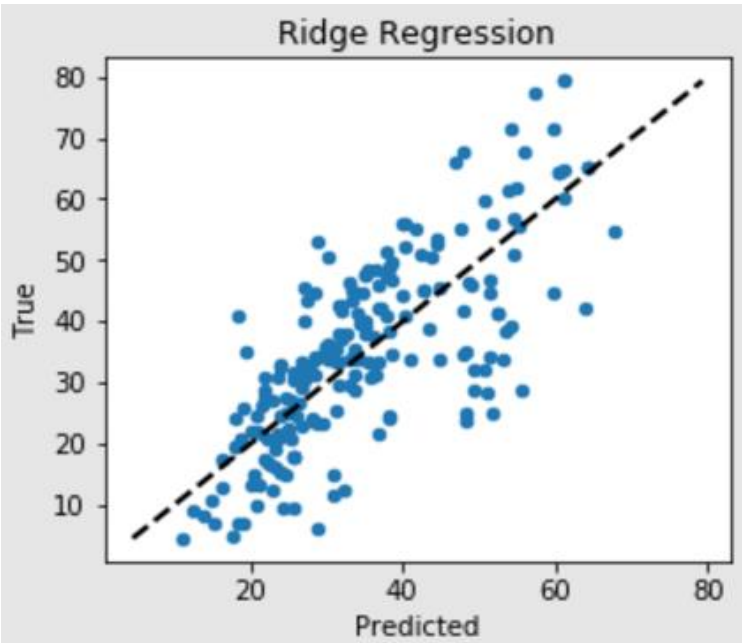

Figure 4.3.1 True vs Predicted values of Ridge Regression

Model	RMSE	MSE	MAE	R^2
Ridge Regression	10.29	105.86	8.24	0.57

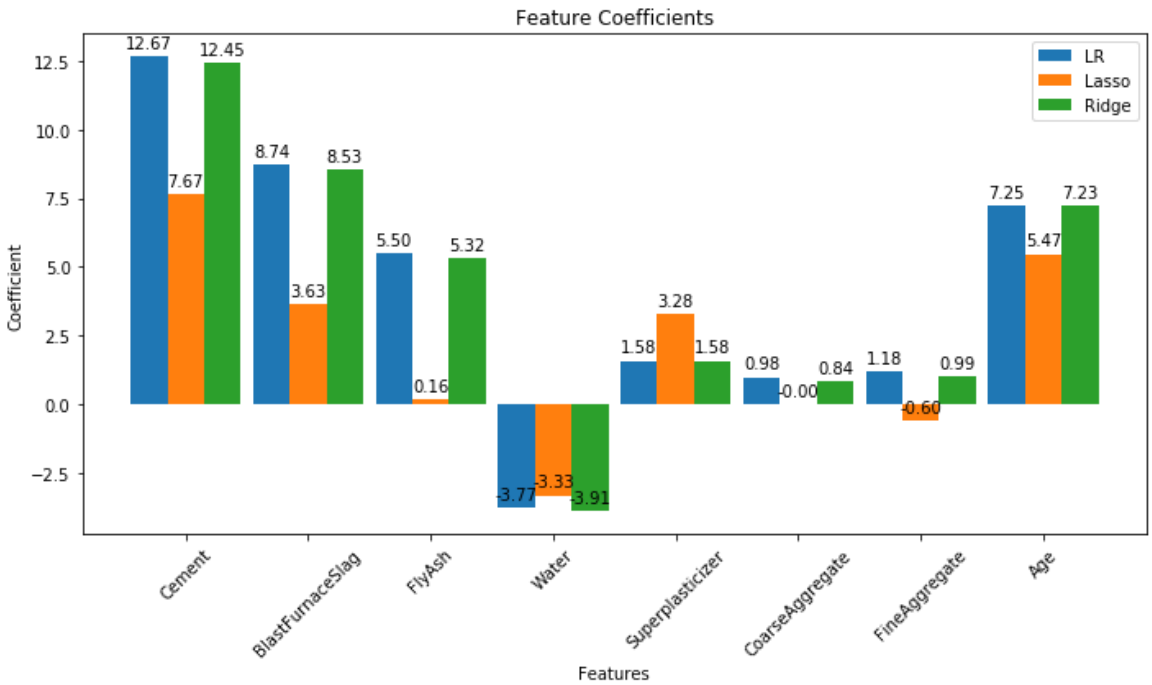

Figure 4.3.2 Feature coefficients for the Linear, Lasso and Ridge Regressions

4.4 Decision Tree Regressor

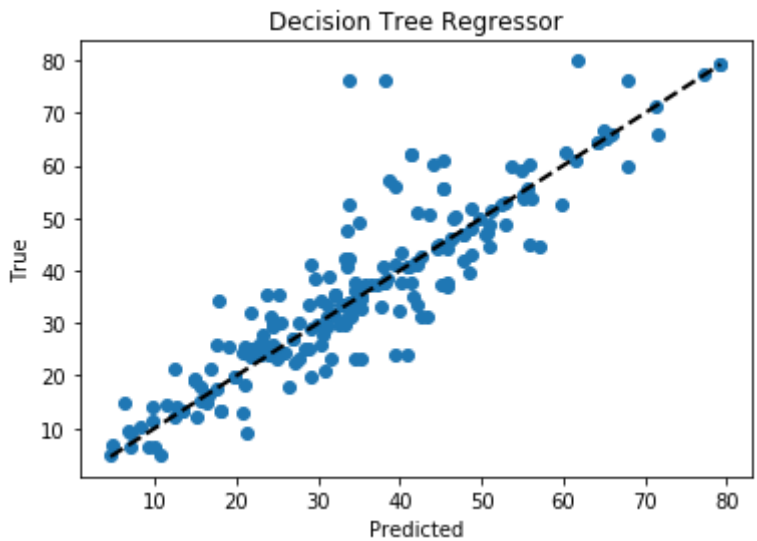

Figure 4.4.1 True vs Predicted values of Decision Tree Regressor

Model	RMSE	MSE	MAE	R^2
Decision Tree Regressor	7.53	56.69	4.66	0.77

4.5 Random Forest Regressor

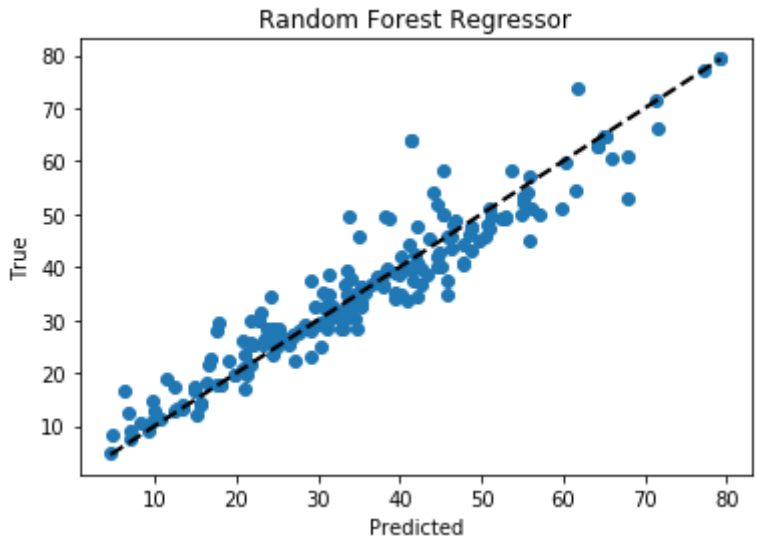

Figure 4.5.1 True vs Predicted values of Random Forest Regressor

Model	RMSE	MSE	MAE	R^2
Random Forest Regressor	5.08	25.85	3.51	0.89

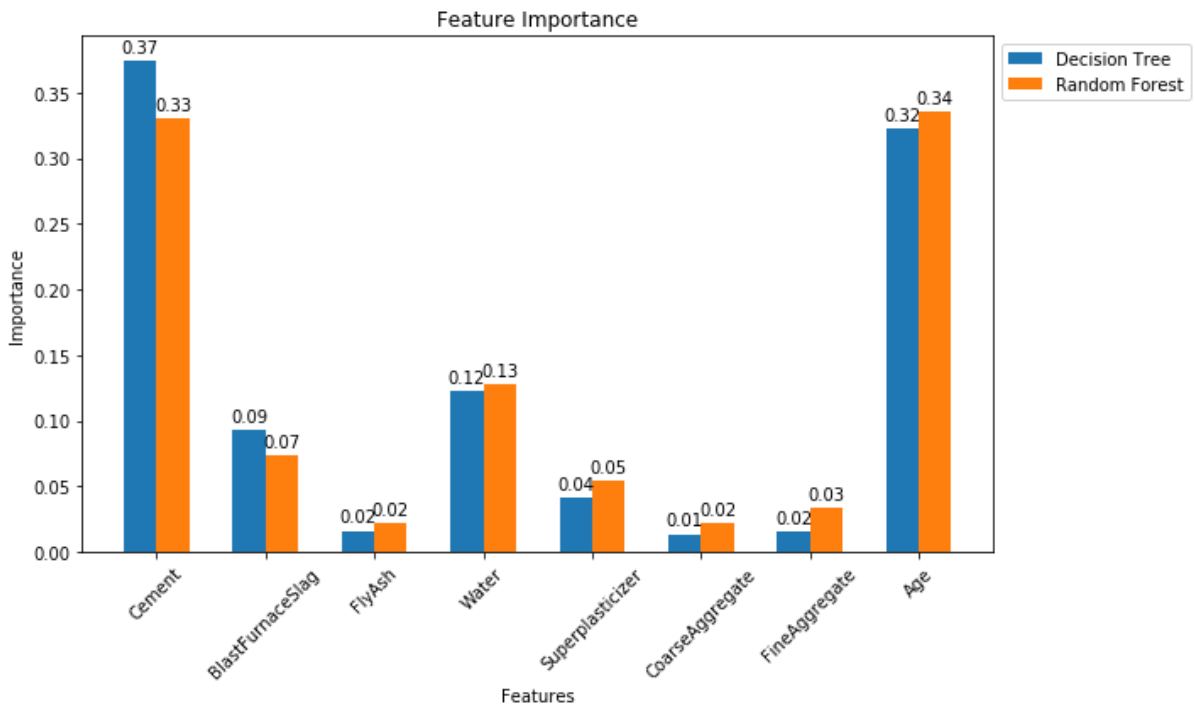

Figure 4.5.2 Importance of individual feature in the strength of concrete

This chart shows the importance of the respective features of the inputs from the predicted outputs by the models.

We can say that Cement and Age of concrete are the most important features which govern the compressive strength of the concrete.

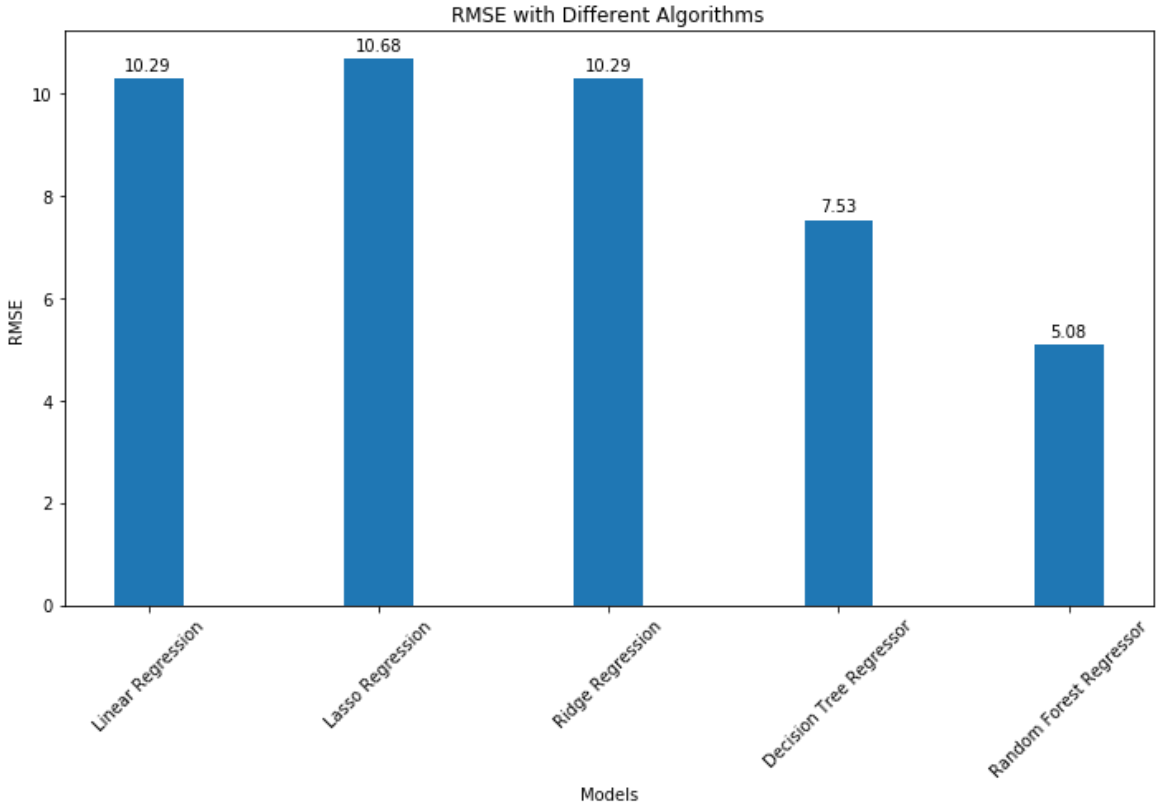

Figure 4.5.3 RMSE of all the models

The formulae for different evaluation metrics are as follows

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

where, y_i represents the actual value,

$\hat{y}_i$ represents the predicted value,

$\bar{y}$ represents the mean of the actual values and n represents the no. of data points

5. CONCLUSIONS

From all the models and their performance, we can say that out of the models that are employed in this Random Forest Regressor is the one that is having the best error evaluation metrics with the least values of RMSE, MSE, MAE and maximum R-squared(R^2) score.

This is because Random Forest uses more deep decision trees and also uses ensemble learning techniques such as bagging. Bagging is an ensemble machine learning technique that improves the accuracy and stability of models by combining the predictions of multiple models. It primarily reduces the variance in the prediction models.

Also from the feature importance and the feature selection charts we can conclude that the cement content and the age of the concrete are the two of the most important features which govern the compressive strength of the concrete. This is because cement plays the role of binding the aggregates. As the cement content increases with proper w/c ratio the concrete has more binding and better hydration which plays a role for concrete in achieving strength.

Whereas, as the age of the concrete increases, the strength also increases. This is due to the fact, after the setting of the concrete for 28 days, it usually achieves 70-80% of its total strength. This is because as the hydration of the concrete increases it becomes denser and stronger. This is why we keep watering the concrete structures over sometime after the construction. As it progresses it achieves 85-90% of strength in the initial 6 months and 95% of the strength in 1 year of the age of concrete.

6. FUTURE SCOPE OF RESEARCH

While my current study focused on predicting the compressive strength of the concrete, there is scope in developing a model which could also give the output as a mix proportion of the ingredients, given the compressive strength of the concrete as the input. There is also scope for developing some more models which could improve the efficiency and accuracy of the predicted values using these models. And there is also a need to create a User Interface (UI) and make these results accessible to everyone.

Whereas the future scope of machine learning for mix design of concrete is very vast. This includes the consideration of more variables into the account for addressing the problem of variability in the predictions, i.e., enhanced accuracy in the prediction. Also durability and long term performance of the concrete can be predicted.

The problems such as sustainability and carbon footprint can also be addressed with the adoption of machine learning in the concrete industry. More optimised solutions can be retrieved using machine learning with large datasets. Also automation in the mix design can be one more important aspect in the future scope of machine learning in the concrete industry, due to more advanced neural networks.

7. REFERENCES

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